University of Science & Technology Chittagong (USTC)

Department of Computer Science & Engineering

Presented By : Pritam Nath Dipta . ID 20010112

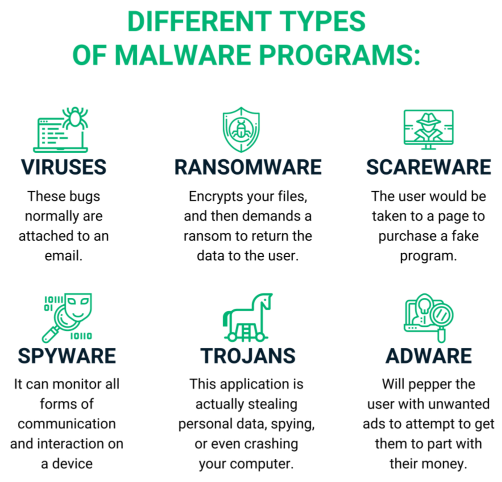
Supervisor :
Nusrat Sharmin,
Lecture, CSE, FSET, USTC

Co-Supervisor :
Topu Biswas,
Research Lecture, CSE, FSET, USTC

Malware Detection Using Machine Learning

Introduction

- ❑ Approximately 5 billion individuals are online worldwide, according to Internet statistics.[2] As a result, everyone is getting different kinds of data from the internet, which makes it very easy for different kinds of hazardous viruses to infiltrate our computers or mobile devices.

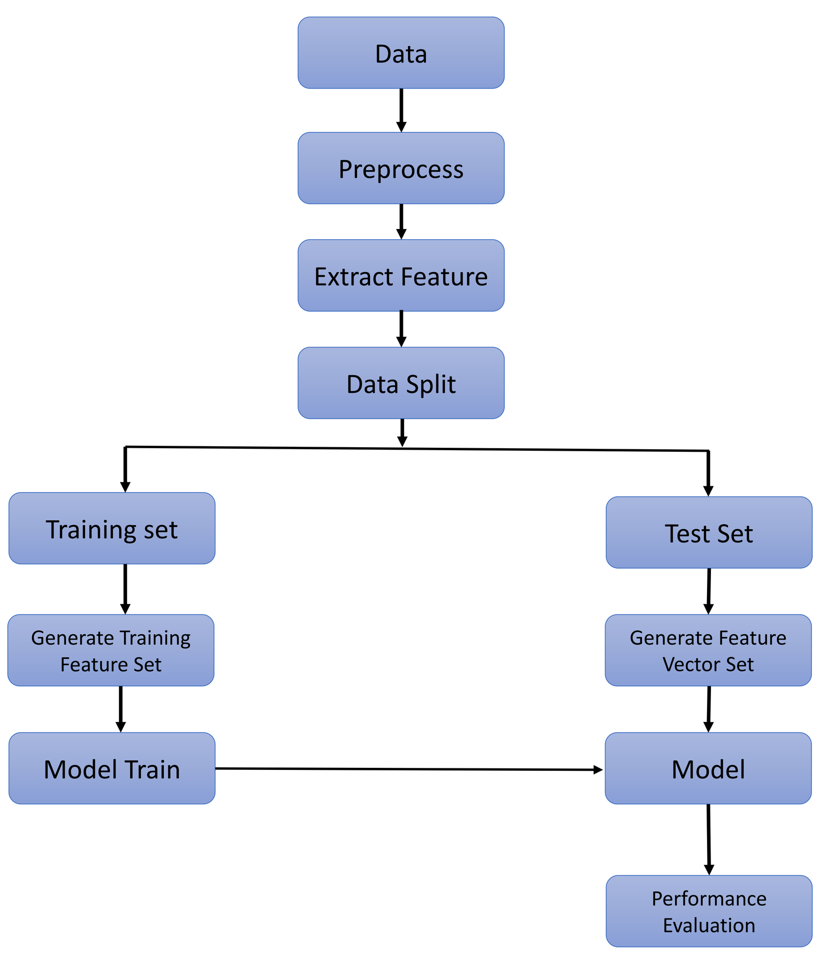


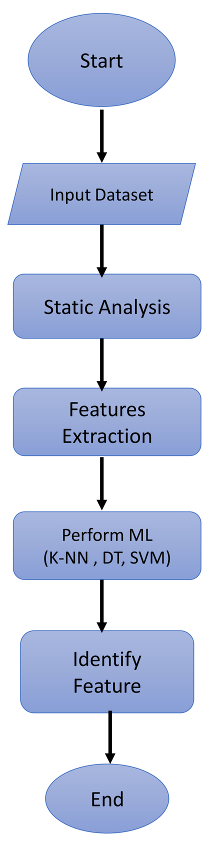
- ❑ One of the key advantages of machine learning-based malware detection is its ability to adapt and learn from new malware samples.[1]
- ❑ Using machine learning, we can recognize the underlying patterns, features, and behaviors that differentiate malware from benign software.

Objective

- ❑ To detect high accuracy, we use machine learning method that can effectively distinguish between malware and benign software. Minimize false positives (classifying legitimate software as malware) and false negatives (failing to detect malware).[3]
- ❑ To develop a model that can detect new and unseen types of malware.
- ❑ To identify features or characteristics of malware that can be used as input to machine learning algorithms.[2]

Methods & Materials





Material :

Problem Statement

- ❑ Feature selection is the main issue here. Effectively separating malware from innocuous software is necessary for reliable detection.
- ❑ Malware authors actively develop evasion techniques to circumvent detection systems.
- ❑ Achieving real-time or near real-time detection of malware is critical, especially in high-security environments

Conclusions

- ❑ We will be used to extract the features and detect the malware by applying machine learning. The results will be compared to increase the dataset to improve the accuracy.
- ❑ For comparison, we use different types of machine learning techniques like KNN, SVM, and DT.
- ❑ In the future, we can use this analytics data to improve our system to protect from malware.

❑ Future Works:

- ❑ Developing algorithms and architectures for real-time or streaming malware detection is crucial in rapidly evolving environments where prompt detection and response are essential.
- ❑ We can develop an AI system that can allow security experts to understand the reasoning behind malware detection and aid in the identification of new malware families.

References

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